

BRE AI

Satin Finserv Limited · Business Rules Engine + AI Underwriting

BRE AI reads an applicant's real financial data, scores them with trained machine-learning models, checks them against the lender's own approval rules, and hands back one clear answer approve, review, or reject every time, consistently.

About the product

Every loan application raises the same question for the credit team:

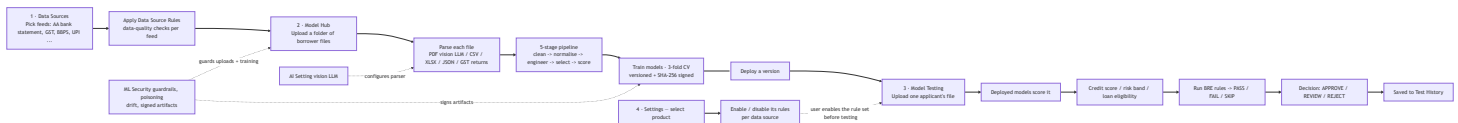
should we approve this, review it, or reject it and for how much?

BRE AI is built to answer that question quickly and consistently, using the applicant's own data rather than guesswork.

It's organised into four simple workspaces:

Workspace	What it's for
Data Sources	Choose which feeds to underwrite on (bank statements, GST filings, UPI, property records, ...) and set the rules for each.
Model Hub	Upload real borrower files; cleaning, training and testing the scoring models is automatic. Every version is kept.
Model Testing	Upload one applicant's file → instant credit score, risk band, and an approve / review / reject decision.
Settings	Rule config pick a product, then turn its approval rules on or off per data source. AI config choose the AI model that reads scanned documents. Plus a view of the security checks.

End-to-end workflow



Step by step

1. Pick the data feeds and set their quality rules Data Sources

- The team picks which of the 11 data feeds this loan product will be underwritten on bank statements, GST filings, utility-bill history, UPI activity, and so on.
- For each feed, the **Data Source Rules** are applied here checks on the incoming data itself (e.g. is the consent valid, are transactions duplicated, is the statement complete and recent enough). These are **data-quality gates**, not underwriting decisions, and they are applied **before training** they decide what data is trustworthy enough to teach the models with.

2. Build the ML Model Hub

- The team uploads a whole folder of real borrower files for the feed they picked in step 1 no single sample file, the actual data.
- Every file is read and cleaned up automatically, whatever format it's in: PDFs, spreadsheets, CSVs, JSON exports, even a photo or scan of a statement (an AI vision model reads those).
- That data is then refined in five simple steps:
 - i. **Clean it up** remove noise and fill in gaps.
 - ii. **Standardise it** put every number on the same scale so nothing unfairly dominates.
 - iii. **Build useful signals** turn raw numbers into meaningful measures, like income stability or turnover trend.
 - iv. **Keep what matters** drop the signals that don't actually help.
 - v. **Produce the final training set** ready to teach the models with.
- Then **picks an algorithm** to train with Gradient Boosting/ XGBoost/ Random Forest a couple of others are available and starts training.
- The models are **trained** on that clean data, and their accuracy is measured honestly through cross-validation real testing, not a guess.
- Every time training runs, the result is saved as a **new version** older versions are never deleted, so the team can always step back to one that

worked.

- Once a version is ready, the team **Upload** it that's what makes it the version Model Testing actually uses to score real applicants.

3. Set the rules — Settings

- **First pick the loan product** (LAP / SBL, Machine Loan, Vehicle Loan, MSME), **then turn its rules on or off** one data source at a time. This is a **second, separate rule book** the one that decides an applicant's outcome (e.g. minimum income, bounce count, GST filing regularity, credit-score threshold), configured per loan product and per data source.
- This is different from the Data Source Rules in step 1: those guard the *training data*; this one is applied to *each applicant* the next time someone tests one in Model Testing.
- Settings is also where the AI model that reads scanned statements is chosen, and where the security checks that guard the data and the models live.
- It's a one-time setup step, done **before** testing begins, and can be adjusted any time.

4. Test an applicant — Model Testing

- The user uploads one applicant's statement (ex. GST filing).
- The deployed model scores it instantly credit score, risk band, projected cash flow, and for GST, loan eligibility and filing-compliance too.
- The rule book configured in Settings is then run against that same applicant, showing exactly which checks passed, which failed, and why.
- All of that combines into one clear outcome: **Approve, Approve with Notes, Conditional Approval, or Reject**. A failure on a critical rule always forces a rejection, so the system never quietly overrides a hard stop.

5. Nothing is lost

- Every test that is run is saved to a history log and reflected on the dashboard, so the team can always look back at what was tested, when, and with what result.